

BIROBIDZHAN, Russian Federation

WMO: 317130

Lat: 48.733N

Lon: 132.950E

Elev: 77

StdP: 100.40

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.3	(c) -32.0	(d) -36.6	(e) 0.1	(f) -33.9	(g) -34.3	(h) 0.1	(i) -31.6	(j) 8.3	(k) -17.3	(l) 6.8	(m) -12.2	(n) 0.3	(o) 270	(p) 0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	29.9	22.6	28.3	21.6	26.7	20.8	24.0	28.3	22.9	26.8	21.9	25.5	2.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.5	26.5	21.5	16.4	25.3	20.6	15.4	24.2	72.8	28.1	68.5	26.8	64.6	25.5	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.2	(b) 6.4	(c) 5.7	(e) -36.8	(f) 32.7	(g) 2.6	(h) 1.9	(i) -38.7	(j) 34.0	(k) -40.2	(l) 35.1	(m) -41.7	(n) 36.2	(o) -43.6	(p) 37.6		
(a) 8.2	(b) 6.4	(c) 5.7	(e) -37.1	(f) 25.5	(g) 2.7	(h) 0.9	(i) -39.0	(j) 26.1	(k) -40.6	(l) 26.7	(m) -42.1	(n) 27.2	(o) -44.0	(p) 27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-21.3	-16.3	-6.8	4.5	12.5	18.2	21.5	19.8	13.5	4.2	-8.7	-19.5		
	DBStd	15.77	4.85	4.87	5.78	4.12	3.85	3.33	2.50	2.79	3.77	4.22	6.48	4.88		
	HDD10.0	4090	969	737	522	171	19	0	0	0	11	184	562	916		
	HDD18.3	6213	1228	970	780	414	183	44	4	16	151	438	812	1174		
	CDD10.0	1131	0	0	0	7	98	245	358	304	115	4	0	0		
	CDD18.3	212	0	0	0	0	3	39	103	62	4	0	0	0		
	CDH23.3	1533	0	0	0	5	69	344	701	376	38	1	0	0		
	CDH26.7	359	0	0	0	1	10	87	193	65	3	0	0	0		
	WSAvg	1.8	1.2	1.5	2.1	2.5	2.4	1.9	1.6	1.6	1.8	2.1	1.9	1.2		
(15) Precipitation	PrecAvg	749	8	11	26	38	76	97	169	148	92	49	20	14		
	PrecMax	1010	22	48	67	97	169	250	373	233	237	142	71	56		
	PrecMin	533	0	0	0	3	13	30	40	28	20	5	0	0		
	PrecStd	133	7	12	20	26	42	63	78	46	55	34	19	14		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.5	0.7	10.5	23.1	28.2	31.6	32.6	30.4	26.7	20.2	8.3	-3.4		
		MCWB	-6.3	-2.9	4.2	11.7	17.2	22.3	24.0	24.1	19.9	12.3	3.6	-4.9		
	2%	DB	-7.8	-2.6	6.4	18.3	25.2	29.0	30.6	28.4	24.1	16.5	5.1	-6.9		
		MCWB	-9.2	-5.2	1.6	9.7	15.2	20.8	23.4	22.5	18.0	9.7	1.7	-8.1		
	5%	DB	-10.9	-5.2	4.1	15.0	22.9	27.1	28.7	26.9	22.3	13.7	2.4	-9.3		
		MCWB	-12.0	-7.1	0.2	7.9	13.8	20.0	22.7	21.5	16.7	7.7	0.0	-10.3		
	10%	DB	-13.2	-7.6	2.1	12.2	20.6	25.1	27.0	25.4	20.4	11.3	0.5	-11.5		
		MCWB	-14.0	-9.1	-0.9	6.2	13.1	18.9	21.8	20.4	15.8	6.6	-1.3	-12.4		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.0	-2.6	4.5	12.9	18.1	23.9	25.6	25.1	21.0	13.4	4.1	-4.5		
		MCDB	-4.7	0.7	9.7	21.4	25.3	30.2	30.5	28.8	25.2	18.1	7.8	-3.4		
	2%	WB	-9.2	-5.0	2.2	10.2	16.4	22.0	24.3	23.7	19.3	10.7	2.0	-7.9		
		MCDB	-7.9	-2.9	5.7	17.1	23.0	27.6	28.8	26.9	22.9	14.5	4.3	-6.9		
	5%	WB	-12.0	-7.0	0.6	8.4	15.1	20.7	23.3	22.5	17.8	8.8	0.4	-10.3		
		MCDB	-10.9	-5.2	3.6	14.2	21.1	25.8	27.6	25.7	20.7	12.6	2.1	-9.3		
	10%	WB	-14.0	-9.0	-0.8	6.7	13.7	19.6	22.4	21.4	16.6	7.1	-1.4	-12.4		
		MCDB	-13.2	-7.6	2.0	11.4	19.1	24.1	26.2	24.3	19.6	10.8	0.5	-11.5		
(35) Mean Daily Temperature Range	MDBR		13.5	14.9	12.4	11.1	11.4	10.2	9.0	9.2	11.6	11.3	10.4	12.0		
	5% DB	MCDBR	13.9	15.8	12.7	16.3	16.2	13.9	12.3	11.5	14.5	15.7	11.5	11.8		
		MCWBR	12.4	13.3	8.6	9.1	7.6	6.9	6.1	6.0	8.0	9.0	8.2	10.5		
	5% WB	MCDBR	13.7	15.4	11.9	14.9	13.9	12.5	10.7	9.8	11.5	14.1	10.8	11.3		
		MCWBR	12.3	13.0	8.1	8.7	7.1	6.7	5.9	5.5	6.9	8.7	8.0	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.322	0.399	0.547	0.573	0.521	0.506	0.446	0.411	0.429	0.354	0.292		
	taud		2.170	2.042	1.889	1.714	1.744	1.964	2.052	2.223	2.228	2.000	2.085	2.136		
	Ebn at Noon		760	802	787	704	709	755	760	793	781	662	642	682		
	Edh at Noon		80	116	162	216	218	176	159	128	116	122	89	73		
(44) All-Sky Solar Radiation	RadAvg		1.86	3.11	4.52	4.76	5.20	5.53	5.09	4.67	3.85	2.62	1.74	1.44		
	RadStd		0.05	0.09	0.34	0.51	0.47	0.78	0.50	0.35	0.37	0.21	0.13	0.08		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page